Warm-up

(1) Give a formula for the inverse of the function $f(x) = x^5 - 2$.

(2) Is $f(x) = x^2 + x$ one-to-one? Justify your answer.

Medium

(1) is the function $f(x) = \sqrt[4]{1 + 2x^3}$ one-to-one? Justify your answer.

(2) Given that $f(x) = 2x + \cos(x)$ is one-to-one, calculate $(f^{-1})'(1)$.

(3) Find f^{-1} where f is the function $\sqrt{x^2 + 2x}$ with domain $[0, \infty)$. Be sure the formula you find gives an invertible function, and explicitly check your work by showing that $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$.

(4) Given that $h(x) = \sqrt{x^3 + x + 6}$ is one-to-one, compute $(h^{-1})'(4)$.

Stretch

(1) Suppose f is invertible, and that f and f^{-1} are differentiable. If f has antiderivative F, check that

$$\int f^{-1}(x)\,dx = xf^{-1}(x) - F(f^{-1}(x)) + C.$$

(2) Suppose f is twice differentiable. Find a formula for $(f^{-1})''(x)$.